Erfan Shahabi

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Research Interests

Natural Language Processing, Large Language Models, Information Retrieval, Machine Behavior, Social and Strategic Behavior of LLM Agents, User Simulation, LLM Evaluation and Benchmarking

Education

M.Sc in Information Technology Engineering, GPA: 4.00/4.00 (17.76/20) *Sep 2024 – Present*
University of Tehran, School of ECE *Tehran, Iran*

- Supervisors: Dr. Azadeh Shakery, Dr. Yadollah Yaghoobzadeh
- Thesis: An Investigation and Enhancement of Strategic Behavior of Large Language Models in Social Context

B.Sc in Industrial Engineering, GPA: 4.00/4.00 (18.07/20) *Sep 2019 – Aug 2023*
Islamic Azad University, Science and Research Branch *Tehran, Iran*

- Supervisor: Dr. Abbas Saghaei
- Final Project: Used Car Price Prediction using Machine Learning Methods

Publications

- **SimInter: Evidence-Grounded User Simulation with Interview-Based Profile Enrichment for Conversational Recommender Systems**
  *Under Review at **NAACL 2027 Conference** (recommended for Findings in ARR May 2026 cycle)* | arXiv:2610.18308
  E Shahabi*, A Roshandel*, MS Sadeghi, A Shakery (*equal contribution, ordered randomly)
- **PerSo: Investigating Facets of LLMs' Social Intelligence in Real-World Persuasion Scenarios**
  *Under Review at **NAACL 2027 Conference*** | arXiv:2610.20548
  A Zamani*, S Esfahanian*, **E Shahabi**, MT Pilehvar, A Shakery, Y Yaghoobzadeh (*equal contribution, ordered randomly)

Professional Experience

Researcher *Sep 2024 – Present*
Intelligent Information Systems Lab *ECE @ University of Tehran, Tehran, Iran*

- Supervisors: Dr. Azadeh Shakery, Dr. Yadollah Yaghoobzadeh
- Research Area: Strategic and Social Behavior of LLMs, Conversational AI
- Built a behavior-grounded user simulation framework (**SimInter**) for conversational recommender systems, leveraging audited synthetic interviews to enrich user profiles
- Developed **PerSo**, a benchmark evaluating social intelligence of LLMs using real-world persuasion scenarios from Reddit's Change My View (CMV) community

AI Engineer / Data Scientist *Jul 2026 – Present*
Potk.ir *Tehran, Iran*

- Built price forecasting pipelines and LLM- and retrieval-based tools for the electrical cable market
- Contributed to turning customers' informal requests into structured data

AI Engineer *Feb 2026 – Jul 2026*
Mofid Stock Brokerage *Tehran, Iran*

- Built LLM-based RAG assistants to teach stock-market concepts to users from novice to professional
- AI consultant to Easy Trader, the country's most widely used trading platform, on its trading assistant

Leadership Experience

Research Supervisor, 2 Undergraduate Students *Mar 2026 – Aug 2026*
Intelligent Information Systems Lab *ECE @ University of Tehran, Tehran, Iran*

- Supervised two high-achieving undergraduate students on **PerSo**, a benchmark evaluating LLM social intelligence in real-world persuasion dialogues
- Resulted in a paper submitted to ACL ARR (October 2026 cycle, NAACL 2027); preprint on arXiv

Teaching Experience

Head Teaching Assistant, Intelligent Information Retrieval (Graduate) *Sep 2026 – Present*
Course Website · *Instructor: Dr. Azadeh Shakery* *ECE @ University of Tehran*

- Coordinated 6 computer assignments, workshops & the paper presentation module with a team of 6 TAs

Teaching Assistant, Natural Language Processing (Graduate) *Sep 2025 – Present*
Course Website · *Instructor: Dr. Heshaam Faili* *ECE @ University of Tehran*

- Designed TravelBot assignment: a conversational travel assistant using ReAct architecture and LangGraph for flight search, hotel booking, and trip planning
- Implemented RAG pipeline for retrieving destination information and generating personalized travel recommendations
- Led workshops mentoring students on LangChain and LangGraph frameworks for building agent-based systems

Teaching Assistant, Intelligent Information Retrieval (Graduate) *Sep 2025 – Sep 2026*
Course Website · *Instructor: Dr. Azadeh Shakery* *ECE @ University of Tehran*

- Designed assignments on conversational search, RAG systems, and multi-agent architectures
- Conducted workshops on probability theory and machine learning foundations for information retrieval
- Led technical workshops on LLMs in IR, covering LangChain, LangGraph, and ReAct patterns
- Guided students in implementing practical retrieval systems using state-of-the-art language models

Teaching Assistant, Advanced Data Mining (Graduate) *Feb 2026 – Present*
Instructor: Dr. Azadeh Shakery *ECE @ University of Tehran*

- Designed GNN-based assignment on academic network analysis, where students predict paper authorship and inter-paper relationships using graph neural networks
- Conducted hands-on workshops on GNN architectures and their applications in network data mining

Teaching Assistant, Natural Language Processing (Graduate) *Feb 2026 – Present*
Instructor: Dr. Yadollah Yaghoobzadeh *ECE @ University of Tehran*

- Designed and managed a financial advisory chatbot assignment, guiding students in building domain-specific conversational agents using LLMs
- Led workshops on fine-tuning and prompt engineering for task-specific NLP applications

Selected Projects

- Fine-Tuning LLaMA-3.2 using ORPO and DPO for Human-Preference-Based Alignment
- Quantization-Aware Reasoning and Text-to-SQL Agent Design using ReAct and LangGraph
- ChatFood: Building a ReAct-based Conversational Food chatbot with LangGraph and Gemini
- Conversational Search with RAG: Dataset Construction and Multifaceted Evaluation
- Deep Learning for Sequence Labeling and Text Summarization with RNNs and Attention
- Sentiment Classification of IMDB Reviews using Text Preprocessing and Classical IR Techniques
- Relevance Classification using BERT for Query-Document Pairs in IR Tasks
- Forecasting Community Dynamics in Social Graphs using Multi-Task GNNs

Technical Skills

- **Soft Skills:** Teamwork, Mentoring & Teaching, Meeting Tight Deadlines, Cross-cultural Communication
- **Programming Languages:** Python
- **Large Language Models:** Hugging Face, LangChain, LangGraph, LangFuse, Unsloth, PEFT, Datasets, TRL
- **Databases:** SQL, LanceDB
- **Machine Learning:** PyTorch, Scikit-Learn, Evaluate, TensorFlow
- **Retrieval:** Sentence Transformers, Lexical Retrieval, Semantic Retrieval, BM25

Honors and Awards

- Ranked 14 in Nationwide Graduate Level University Entrance Test of Information Technology Engineering

Languages

- **Persian:** Native **English:** IELTS Academic 7/9 (CEFR C1)